

Pyrimidine 5'-Nucleotidase Deficiency (P5N / UMPH1 deficiency)

Comprehensive Disease Characterization Report

MONDO:0009946 · OMIM #266120 · Orphanet ORPHA:35065

Evidence base: This report is compiled from primary literature (case series, molecular characterization studies, biochemical/enzymology reviews) identified via PubMed. Evidence is predominantly **human clinical/biochemical**, with supporting **in vitro** enzymology and historical **animal (rabbit) lead-poisoning models**. Individual-patient data are drawn from case reports; disease-level facts from OMIM/Orphanet/review syntheses. No primary datasets were provided; findings are literature-derived.

1. Disease Information

Overview. Pyrimidine 5'-nucleotidase (P5N) deficiency is a rare, inherited red-blood-cell enzymopathy causing **chronic hereditary nonspherocytic hemolytic anemia (HNSHA)**. The deficient enzyme — cytosolic **pyrimidine 5'-nucleotidase type I (P5'N-1 / cN-III)** — normally dephosphorylates pyrimidine 5'-ribonucleotides (UMP, CMP) that are generated when the maturing reticulocyte degrades its ribosomal RNA. When the enzyme is absent, these pyrimidine nucleotides cannot be dephosphorylated to diffusible nucleosides and accumulate inside the erythrocyte, producing the disease's diagnostic hallmark: **marked/coarse basophilic stippling** on the blood film together with a shortened red-cell lifespan (

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It is regarded as one of the more common HNSHA-causing enzymopathies — the **third most frequent red-cell enzyme defect causing hemolysis after G6PD and pyruvate kinase**

deficiency

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Key identifiers. - **OMIM (phenotype):** #266120 — "Anemia, hemolytic, due to UMPH1 deficiency" - **OMIM (gene):** 606224 (*NT5C3A*) - *Orphanet:* ORPHA:35065 ("Pyrimidine 5'-nucleotidase deficiency") - *MONDO:* MONDO:0009946 - *ICD-10:* D55.3 (anemia due to disorders of nucleotide metabolism) / D55.8 - *ICD-11:* 3A11.Y (other specified enzyme deficiency anaemias) - *MeSH:* related terms "Anemia, Hemolytic, Congenital Nonspherocytic"; "Pyrimidine Nucleotidase"; "5'-Nucleotidase" - *HGNC gene:** NT5C3A (HGNC:17820)

Synonyms / alternative names. - Pyrimidine 5'-nucleotidase type I (P5'N-1 / P5N-1 / PN-I / P5NI) deficiency - UMPH1 deficiency (uridine 5'-monophosphate hydrolase 1) - Uridine monophosphate hydrolase deficiency - Cytosolic 5'-nucleotidase III (cN-III) deficiency - Hemolytic anemia due to pyrimidine 5'-nucleotidase deficiency - Historic: "hereditary hemolytic anemia with high red-cell pyrimidine nucleotides"

Data source type. Individual-patient (case reports/EHR-style) for phenotype and variant data; aggregated disease-level resources (OMIM, Orphanet) for definitions and epidemiology.

2. Etiology

Primary cause (genetic). Biallelic loss-of-function variants in **NT5C3A** (formerly NT5C3, P5N-1, UMPH1) encoding cytosolic pyrimidine 5'-nucleotidase-I. Inheritance is **autosomal recessive**

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Disease arises when both alleles are non-functional (homozygous or compound heterozygous).

Genetic risk factors. - *Causal variants:* missense, nonsense, frameshift, and splice-site variants in NT5C3A (see §4). No common susceptibility loci — this is a Mendelian, single-gene disorder. - *Modifier genes:* co-inherited **UGT1A1** promoter (Gilbert TA7 allele) worsens

hyperbilirubinemia/cholestasis and gallstone/iron-overload risk; unstable hemoglobins (**Hb E**) and **α -thalassemia** interact to increase hemolytic severity

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- *Consanguinity*: a major contributor because the disorder is recessive and rare; many reported families are consanguineous

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Environmental risk factor (acquired phenocopy). Lead poisoning inhibits erythrocyte P5N and reproduces an essentially identical syndrome (anemia, basophilic stippling, pyrimidine nucleotide accumulation) — the principal non-genetic cause of the same biochemical phenotype

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Occupational/industrial lead exposure and non-industrial sources (lead-glazed pottery, contaminated home-made wine/spirits, lead plumbing) are relevant exposures

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Protective factors. No specific genetic or dietary protective alleles are established. Avoidance of lead exposure is protective against the acquired phenocopy. Because the anemia is usually mild-to-moderate and compensated, general good iron/health status and avoidance of additional oxidative stressors mitigate severity.

Gene–environment interactions. Lead and the hereditary deficiency converge on the same enzyme; a genetically borderline individual plus lead exposure could show additive enzyme suppression. Modifier alleles (UGT1A1, Hb E, thalassemia) modulate expressivity (epistasis;

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3. Phenotypes

Phenotype types: **laboratory abnormalities** and **clinical signs/symptoms** of chronic hemolysis. Onset is typically **neonatal to early childhood**, though mild cases are diagnosed in adulthood; course is **chronic/lifelong**, usually stable but with hemolytic crises possible.

Phenotype	Type	Onset / severity / frequency	HPO suggestion
Chronic hemolytic anemia (Hb typically 8–11 g/dL)	Lab/sign	Neonatal–childhood; mild–moderate; near-universal	HP:0004870 (nonspherocytic hemolytic anemia); HP:0001878 (hemolytic anemia)
Marked basophilic stippling of erythrocytes	Lab	Present throughout; hallmark; ~universal	HP:0011273 (basophilic stippling of erythrocytes)
Reticulocytosis	Lab	Compensatory; common (e.g., ~7%)	HP:0001923 (reticulocytosis)
Jaundice / unconjugated hyperbilirubinemia	Sign	Childhood; common	HP:0000952 (jaundice); HP:0002904 (hyperbilirubinemia)
Splenomegaly	Sign	Childhood–adult; common	HP:0001744 (splenomegaly)
Cholelithiasis (pigment gallstones)	Sign	Adolescence/ adulthood; common, modifier-dependent	HP:0001081 (cholelithiasis)
Elevated LDH; increased indirect bilirubin	Lab	Chronic; common	HP:0025435 (increased LDH)
Pyrimidine nucleotide accumulation / low purine:pyrimidine ratio	Lab	Constant biochemical marker	(no dedicated HP; laboratory)
Iron overload / raised ferritin (esp. post-splenectomy)	Lab/sign	Adult; variable	HP:0011031 (abnormal iron homeostasis); HP:0040130 (increased ferritin)
Possible learning difficulties (reported, uncertain)	Behavioral	Variable; not established	HP:0001328 (specific learning disability) — tentative

Note on neurocognitive association: The original P5N-1 gene-cloning paper stated the deficiency "is implicated in the anemia of lead poisoning and is **possibly associated with learning difficulties**"

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This link is **unconfirmed** — it likely reflects the confounding of the lead-poisoning phenocopy (lead itself is neurotoxic) rather than a proven effect of the hereditary enzyme defect on the CNS. No controlled neurocognitive data in genetically confirmed hereditary P5N deficiency are available (evidence gap).

Severity/progression. Generally **mild-to-moderate, well-compensated hemolysis**; most patients are not transfusion-dependent. Severity is **variable** and can be aggravated by co-inherited modifiers (Hb E, thalassemia, UGT1A1)

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Basophilic stippling is unusually coarse and abundant, a distinguishing feature versus other HNSHAs.

Quality-of-life impact. Chronic anemia-related fatigue, jaundice, gallstone morbidity (possible cholecystectomy), and — where splenectomy is done — thrombotic risk. Standardized QoL instrument data specific to P5N deficiency are not available (evidence gap).

4. Genetic / Molecular Information

Causal gene. **NT5C3A** (aliases NT5C3, P5N-1, UMPH1, PN-I, cN-III; HGNC:17820; NCBI Gene 51251; Ensembl ENSG00000122643; **UniProt Q9H0P0**), **chromosome 7p14.3** (mapped 7p15–p14). Structure: **10 exons**, alternative splicing of exon 2, producing protein isoforms of **286 and 297 amino acids**

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DNA analysis is complicated by **P5'N-1 pseudogenes on chromosomes 4 and 7** (P 11369620).

Protein family: InterPro HAD-like hydrolase / Pfam PF05822 (5'-nucleotidase family).

Enzyme. Cytosolic 5'-nucleotidase type III (cN-III / P5'N-1; EC 3.1.3.5), a member of the **haloacid dehalogenase (HAD) superfamily**; it has both hydrolytic (pyrimidine 5'-monophosphate → nucleoside + Pi) and phosphotransferase activities and requires Mg^{2+} (P 23992312).

Recombinant human P5'N-1 is **a relatively stable protein with essentially identical catalytic efficiency toward CMP and UMP** (its two physiological substrates) (P 15604219).

Representative pathogenic variants (all germline; loss-of-function). | Variant (nomenclature as reported) | Type | Population | Ref | |---|---|---|---| | c.693+1G>A (splice) | Splice-site | LoF | | Turkish |

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| | c.393_394delTA (frameshift) | Frameshift | LoF | | Turkish |

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| | p.R56G (c.166C>G) "Campinas" | Missense (conserved) | Brazilian/African descent |

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| | p.Asp98Val (codon 98 GAT→GTT) | Missense | — |

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| | p.Gln177Ter (CAA→TAA) | Nonsense | — |

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| | IVS9-1 G>T / IVS9-1 g>c (loss of exon 9) | Splice-site | — |

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| | p.Asn190Ser (AAT→AGT) | Missense | Italian |

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| | DelG576, InsGG743 (frameshift) | Frameshift LoF | Southern Italian / Turkish |

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| | 543T>G (Tyr181Ter) | Nonsense | Turkish |

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| | 384-385insA | Frameshift | Turkish |

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Variant classification / functional consequence. Reported variants are **pathogenic/likely pathogenic (ACMG/AMP)** and act via **loss of enzyme function** (reduced catalytic activity; enzyme activity often ~10–30% of normal, e.g., 15%). Frameshift/nonsense/ splice variants truncate or abolish protein; missense variants (R56G, D98V, N190S) affect conserved catalytic/structural residues

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Founder/geographic clustering: DelG576/InsGG743 in southern Italy; 743-744insGG recurrent in Turkish families

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Functional studies (in vitro, PS3-type evidence). Recombinant expression of missense mutants **D87V, L131P, N179S, and G230R** demonstrated that all "display impaired catalytic properties and/or reduced thermostability," and that mutations "affect amino acid residues unambiguously essential for the catalytic efficiency and/or protein stability"

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~15 distinct pathogenic mutations were catalogued by 2006. Notably, **there is NO correlation**

between residual enzyme activity and degree of hemolysis, and some patients retain moderate RBC activity — implying **compensation by other nucleotidases/alternative nucleotide pathways**; thus nucleotidase activity is not a reliable prognostic indicator

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Allele frequency. Individually very rare in gnomAD/population databases (consistent with an ultra-rare recessive disorder); most are private/founder variants. Somatic origin — not applicable (constitutional germline).

Modifier genes. UGT1A1 (Gilbert), HBB (Hb E), α -globin (HBA1/HBA2, α -thalassemia) — modify severity

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Epigenetic / chromosomal abnormalities. No disease-specific methylation or large-scale chromosomal changes are described; disease is caused by point-level NT5C3A lesions.

5. Environmental Information

- **Environmental / toxic factor: Lead (Pb^{2+})** — direct enzyme inhibitor producing an acquired phenocopy (basophilic stippling, pyrimidine accumulation, hemolytic/hypoproliferative anemia)

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CHEBI:25016 (lead).

- **Lifestyle / exposure sources of lead:** occupational (industry, smelting, battery, paint), and non-occupational — lead-contaminated food/water, lead plumbing, artisanal wine/spirits,

lead-glazed ceramics

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- **Infectious agents:** Not applicable — no infectious etiology.

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function NT5C3A variants** (or, in the acquired form, **lead inhibition**) → **abolish cytosolic pyrimidine 5'-nucleotidase (P5'N-1/cN-III) activity** (demonstrated;

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2. Loss of P5'N-1 → **failure to dephosphorylate pyrimidine 5'-ribonucleotides (UMP, CMP)** that are produced when the **maturing reticulocyte degrades ribosomal RNA** (demonstrated;

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3. Because 5'-monophosphates (unlike nucleosides) **cannot cross the RBC membrane**, undegraded **pyrimidine nucleotides accumulate intracellularly** (7–80% of nucleotide pool in the lead phenocopy) (demonstrated;

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4. Accumulated **ribonucleotides/ribonucleoprotein aggregate** → visible as **coarse basophilic stippling** on the stained film (demonstrated correlate;

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5. Branch A — **metabolic inhibition:** high pyrimidine nucleotides act as **metabolic inhibitors**, and enzyme-deficient young cells show **inhibition of the hexose-monophosphate (pentose phosphate) shunt** → reduced NADPH/glutathione antioxidant

capacity → **oxidative denaturation of hemoglobin** (demonstrated in Hb E co-inheritance;
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6. Branch B — **nucleotide-pool distortion / energy metabolism**: abnormal pyrimidine nucleotides may **compete with adenine nucleotides** and perturb ATP-dependent processes and pyrimidine-dependent membrane-lipid synthesis (e.g., CDP-choline pathway) — *inferred*, exact step "still unclear"/"not elucidated"

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7. Convergence → **membrane/oxidative injury and reduced deformability** → **premature erythrocyte destruction (extravascular hemolysis, splenic)** → **chronic hemolytic anemia** (demonstrated phenotype;

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8. Downstream sequelae → **reticulocytosis, unconjugated hyperbilirubinemia, jaundice, splenomegaly, pigment gallstones, and (variably) iron overload** (demonstrated;

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The precise biochemical step that shortens red-cell survival remains **incompletely defined** ("mechanism of still unclear mechanism";

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"the precise metabolic process ... has not been elucidated yet",

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— the strongest current evidence favors combined ribonucleoprotein burden plus HMP-shunt/antioxidant impairment.

Category checklist

- **Molecular pathways**: pyrimidine nucleotide catabolism/salvage; pentose phosphate (HMP) shunt; glutathione redox. KEGG pyrimidine metabolism (hsa00240).
- **Cellular processes**: ribosomal RNA turnover during reticulocyte maturation; oxidative stress response; premature erythrophagocytosis. GO:0006206 (pyrimidine nucleobase

metabolic process); GO:0009117 (nucleotide metabolic process); GO:0006749 (glutathione metabolic process); GO:0034101 (erythrocyte homeostasis).

- **Protein dysfunction:** loss of function of P5'N-1 (HAD-superfamily hydrolase); truncation/destabilization/active-site disruption. GO:0008253 (5'-nucleotidase activity); GO:0002100 (tRNA/nucleotide dephosphorylation-related).
- **Metabolic changes:** intracellular pyrimidine ribonucleotide (UMP/CMP/CDP-derivatives) accumulation; low purine:pyrimidine ratio; reduced NADPH regeneration.
- **Immune involvement:** none primary (non-immune hemolysis; Coombs-negative).
- **Tissue damage mechanism:** oxidative membrane/hemoglobin damage, reduced deformability, splenic sequestration.
- **Biochemical abnormality:** enzyme deficiency (EC 3.1.3.5), pyrimidine 5'-nucleotide accumulation. CHEBI:17568 (UMP-related), CHEBI:17361 (CMP-related).

7. Anatomical Structures Affected

- **Organ / system level:** **Hematopoietic/erythroid system** primary; **spleen** (splenomegaly, sequestration) and **hepatobiliary system** (jaundice, pigment gallstones, cholestasis in modifier cases) secondary; **liver** iron deposition/siderosis in some (P 25153905).
- Body system: **cardiovascular/hematologic**. UBERON:0000178 (blood); UBERON:0002106 (spleen); UBERON:0002107 (liver); UBERON:0002110 (gallbladder); UBERON:0002371 (bone marrow).
- **Tissue / cell level:** **Erythrocytes / reticulocytes** are the target cell population; erythroid precursors in marrow. CL:0000232 (erythrocyte); CL:0000558 (reticulocyte); CL:0000764 (erythroid lineage cell).
- **Subcellular level:** **Cytosol** (site of P5'N-1 and pyrimidine nucleotide accumulation); ribosomes/ribonucleoprotein aggregates (basophilic stippling). GO:0005829 (cytosol); GO:0005840 (ribosome).
- **Localization / lateralization:** systemic (circulating red cells); splenomegaly and organ effects are typically **bilateral/systemic**, not lateralized.

8. Temporal Development

- **Onset:** usually **neonatal to early childhood** hemolytic anemia/jaundice; mild cases may present in adolescence or adulthood (including incidentally). Pattern **chronic/insidious** (

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- **Progression:** **stable chronic** hemolysis in most; not typically progressive. Complications accrue over time (gallstones, iron overload). Hemolytic exacerbations can be **episodic** with intercurrent stress.
- **Disease course / duration:** **lifelong**. No spontaneous remission (enzyme defect is constitutional). "Remission" of the acquired lead phenocopy follows removal of exposure and chelation.
- **Critical periods:** neonatal jaundice window (kernicterus risk if severe, especially with UGT1A1 modifier); lead-exposure windows for the acquired form.

9. Inheritance and Population

- **Epidemiology:** **rare**; exact prevalence not precisely established (Orphanet: <1/1,000,000 order; "ultrarare-rare"). It is the **third most common erythrocyte enzyme abnormality causing hereditary nonspherocytic hemolytic anemia, after G6PD and pyruvate kinase deficiency** (

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In nationwide HHA cohorts, RBC enzymopathies as a class account for only ~6–13% of hereditary hemolytic anemia cases (e.g., Korea 6.2–13.3%), of which P5N is a small fraction (

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~60+ families reported worldwide.

- **Inheritance: Autosomal recessive**

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- **Penetrance/expressivity:** biochemical penetrance essentially complete in homozygotes; **clinical expressivity variable**, modifier-dependent

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- **Genetic anticipation:** not applicable (no repeat expansion).

- **Germline mosaicism:** not described.

- **Founder effects / geographic clustering:** recurrent variants in **southern Italian** (DelG576) and **Turkish** (743insGG) populations; reported across Mediterranean, Middle Eastern, South Asian, East Asian, and South American (first in Brazil) populations

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- **Consanguinity:** important contributor; many homozygous cases from consanguineous unions

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- **Carrier frequency:** not well quantified; heterozygous carriers are asymptomatic with ~50% enzyme activity.

- **Sex ratio:** autosomal — **no sex predilection** (M:F $\approx$ 1:1).

10. Diagnostics

Clinical/laboratory tests. - **Peripheral blood film:** the key clue — **coarse basophilic stippling** with polychromasia, anisopoikilocytosis; can be recognized from the film alone

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- **Hemolysis panel:** ↓ Hb, ↑ reticulocytes, ↑ LDH, ↑ indirect bilirubin, ↓ haptoglobin; **Coombs-negative** (non-immune). Normal osmotic fragility and normal G6PD (helps exclude membranopathy/G6PD)

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- **Biochemical confirmation: erythrocyte P5N enzyme assay** (activity typically ~10–30% of normal) and demonstration of **elevated intra-erythrocytic pyrimidine nucleotides** — e.g., UV spectral scan of a nucleotide extract showing a shifted 260/280 absorbance ratio, or a **decreased purine:pyrimidine ratio** (1.07 vs 1.4–2.98)

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LOINC: use nucleotidase enzyme activity and RBC count/retic panels. - **Blood lead level:** mandatory to exclude the acquired phenocopy

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Genetic testing. - **NT5C3A single-gene sequencing** or **HNSHA/hemolytic anemia gene panels** (NGS) are the definitive molecular test; **WES/WGS** used when panels are negative. Pseudogenes on chr 4/7 require careful primer/analysis design

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GTR panels for "hereditary hemolytic anemia" include NT5C3A. - Chromosomal microarray/karyotype/FISH: not indicated (point mutations). mtDNA/repeat-expansion testing: not applicable.

Omics-based diagnostics. Metabolomic profiling of RBC nucleotides (pyrimidine accumulation) is diagnostic in research settings; not routine.

Clinical criteria / differential diagnosis. Diagnosis rests on HNSHA + basophilic stippling + enzyme/nucleotide/genetic confirmation. **Differential:** lead poisoning (check Pb, ALAD), thalassemia and hemoglobinopathies (basophilic stippling also seen), G6PD deficiency, pyruvate kinase deficiency, other HNSHA enzymopathies, hereditary spherocytosis (osmotic fragility), sideroblastic anemias

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Screening. No population newborn screening. Cascade/carrier testing offered in affected families; consider in consanguineous pedigrees.

11. Outcome / Prognosis

- **Survival/life expectancy:** generally **good; near-normal life expectancy**. Anemia is usually mild-to-moderate and compensated; rarely transfusion-dependent

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- **Morbidity:** chronic anemia/fatigue, jaundice, **pigment gallstones** (may need cholecystectomy), **iron overload/siderosis** (especially post-splenectomy or with UGT1A1 modifier), and, if splenectomized, **thrombotic complications**

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- **Complications:** neonatal hyperbilirubinemia; gallstones; iron overload; post-splenectomy extreme thrombocytosis and portosplenomesenteric vein thrombosis

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- **Prognostic factors:** co-inherited modifiers (UGT1A1, Hb E, α -thalassemia) predict more severe disease/cholestasis/iron overload

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baseline Hb and transfusion need. **Residual enzyme activity is NOT prognostic** — no correlation exists between residual P5'N-1 activity and degree of hemolysis, likely due to metabolic compensation by other nucleotidases

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- **Recovery:** the acquired lead phenocopy is reversible with exposure removal + chelation; the hereditary form is lifelong.

12. Treatment

No curative pharmacotherapy exists; management is supportive

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"No specific therapy for P5'N-1 deficiency is now available"

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- **Supportive care (NCIT: Supportive Care):** folic acid supplementation, transfusion during crises/severe anemia, monitoring and treatment of gallstones, and **iron-overload surveillance with chelation** when indicated

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NCIT terms: C15277 (Blood Transfusion), C1734 (Deferoxamine)/iron chelation, C542 (Cholecystectomy) for gallstones. - **Splenectomy (NCIT: C15355): generally ineffective** for the anemia and carries significant thrombotic risk — reserved, if ever, for selected refractory cases; a reported case developed extreme thrombocytosis and extensive venous thrombosis post-splenectomy

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- **Lead-poisoning (acquired) form:** remove exposure and administer **chelation** (e.g., calcium disodium EDTA)

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NCIT: C61815 (Edetate Calcium Disodium). - **Advanced/experimental:** red-cell enzymopathies are in principle amenable to **hematopoietic stem-cell transplantation and gene therapy/gene editing**, but none is established specifically for P5N deficiency

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No approved targeted, RNA-based, or cell therapies. - **Pharmacogenomics:** UGT1A1 genotype relevant for bilirubin handling and drug glucuronidation considerations.

13. Prevention

- **Primary prevention:** for the **acquired form**, eliminate/limit **lead exposure** (industrial hygiene, safe water/plumbing, avoiding lead-glazed ware and artisanal lead-contaminated beverages) — CDC/WHO lead-control measures

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No primary prevention for the hereditary form beyond reproductive counseling.

- **Secondary prevention:** early recognition via blood film in a patient with HNSHA; monitoring for gallstones and iron overload for early intervention.
- **Genetic screening/counseling:** **genetic counseling** for autosomal recessive recurrence risk (25% per pregnancy for carrier couples); **cascade carrier testing** and, where desired, **prenatal/preimplantation genetic testing** in families with known NT5C3A variants; especially relevant in **consanguineous** families

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- **Tertiary prevention:** avoid unnecessary splenectomy (thrombotic risk); manage iron overload and gallstones; monitor neonates for kernicterus risk when UGT1A1 modifier present.

- **Immunization / prophylaxis:** if splenectomy is performed, standard **asplenia vaccination** (pneumococcal, Hib, meningococcal) and **antibiotic prophylaxis** apply.

14. Other Species / Natural Disease

- **Taxonomy:** primary human disease. *Homo sapiens* NCBI:txid9606.
- **Orthologous genes:** NT5C3A orthologs exist across mammals (mouse *Nt5c3*, NCBI Gene) and are broadly conserved (the enzyme is ubiquitously distributed in mammalian tissues; [P 23992312](#)).
- **Natural animal disease:** no well-characterized spontaneous P5N-deficiency hemolytic disease in companion animals is established; not a recognized OMIA entry of major veterinary importance.
- **Experimental cross-species relevance:** **rabbit** lead-poisoning models reproduce partial P5N inhibition with basophilic stippling, informing the acquired mechanism ([P 231420](#)).
- **Comparative biology:** enzyme function (pyrimidine nucleotide dephosphorylation during reticulocyte RNA turnover) is evolutionarily conserved, supporting mechanistic translation.
- **Zoonotic potential:** none (non-infectious).

15. Model Organisms

- **Model types available:**
- **In vitro / biochemical:** recombinant human cN-III/P5'N-1 enzymology (substrate specificity, hydrolase + phosphotransferase activities, HAD-superfamily mechanism) — the best-developed model system for structure–function ([P 23992312](#)).
- Recombinant expression of patient missense mutants (D87V, L131P, N179S, G230R) enabled functional dissection of catalytic-efficiency vs thermostability defects ([P 15604219](#)).

- **Animal (induced/toxicological): rabbit** intravenous lead-acetate model producing anemia, microspherocytosis, and basophilic stippling with P5N effects

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- **Cellular:** patient-derived erythrocytes/reticulocytes used to demonstrate nucleotide accumulation and HMP-shunt inhibition

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- **Genetic models:** a dedicated *Nt5c3a* knockout mouse recapitulating the human hemolytic phenotype is **not prominently established** in the literature reviewed (evidence gap); murine ortholog exists for future engineering (MGI).

- **Phenotype recapitulation / limitations:** biochemical/enzymology and toxicological (lead) models reproduce the **pyrimidine-accumulation and stippling** features well, but a genetic animal model fully reproducing chronic hereditary hemolysis is lacking; species differences in reticulocyte RNA content and lead sensitivity limit direct extrapolation

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- **Applications:** enzyme structure–function, substrate specificity, inhibitor (lead) studies, and mechanism of pyrimidine-nucleotide toxicity.

- **Resources:** MGI (mouse *Nt5c3*), UniProt/PDB for enzyme structure, NCBI Gene for orthologs.

Supported vs Refuted Hypotheses

Supported: - P5N deficiency is an autosomal recessive NT5C3A loss-of-function disorder causing HNSHA with basophilic stippling and pyrimidine nucleotide accumulation

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- Lead poisoning is an environmental phenocopy via enzyme inhibition

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 - Hemolysis involves HMP-shunt inhibition / oxidant susceptibility in addition to nucleotide accumulation
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 - Clinical expression is modified by UGT1A1, Hb E, and thalassemia
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 ([P 8839873](#))
 ([P 23384910](#))
 - Splenectomy is generally ineffective and thrombogenic; care is supportive
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Refuted / not supported: - That splenectomy corrects the anemia (it does not;
 ([P 8375297](#))
 - That the disorder is immune-mediated (it is Coombs-negative, intrinsic;
 ([P 39967523](#))
Uncertain / evidence gaps: - The exact biochemical step causing shortened RBC survival remains unresolved
 ([P 23992312](#))
 ([P 2558262](#))
 - Association with learning difficulties is proposed but unproven
 ([P 11369620](#))
 - Precise prevalence, carrier frequency, and a definitive genetic animal model are lacking.

Limitations & Future Directions

- Evidence derives from case reports/small series and biochemical reviews; no large registries give precise epidemiology or QoL metrics.
 - The downstream mechanism of hemolysis is incompletely defined — targeted metabolomic/redox studies in patient reticulocytes are warranted.
 - A faithful *Nt5c3a*-null animal model and structural (PDB/AlphaFold) genotype–activity maps would clarify variant pathogenicity and enable therapy testing (HSCT/gene editing).
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Key Ontology Term Summary

- **MONDO:** MONDO:0009946
 - **Gene (HGNC):** NT5C3A (HGNC:17820)
 - **HPO:** HP:0004870, HP:0011273, HP:0001923, HP:0000952, HP:0001744, HP:0001081, HP:0002904
 - **GO:** GO:0008253 (5'-nucleotidase activity), GO:0006206 (pyrimidine nucleobase metabolism), GO:0034101 (erythrocyte homeostasis), GO:0006749 (glutathione metabolism), GO:0005829 (cytosol)
 - **CL:** CL:0000232 (erythrocyte), CL:0000558 (reticulocyte)
 - **UBERON:** UBERON:0000178 (blood), UBERON:0002106 (spleen), UBERON:0002107 (liver)
 - **CHEBI:** CHEBI:25016 (lead), pyrimidine nucleotides (UMP/CMP)
 - **NCIT:** C15355 (Splenectomy), C15277 (Blood Transfusion), C61815 (Edetate Calcium Disodium)
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